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C Programing Assingnment 1

1. Arithmetic Operators + - * / %

// 1. Write a C program to input two numbers and print their sum, difference, product, quotient, and remainder.

```
#include <stdio.h>
int main(){
    int a, b;
    printf("Enter A and B: ");
    scanf("%d %d", &a, &b);

    printf("\nSum is: %d", a+b);
    printf("\nSub is: %d", a-b);
    printf("\nMulti is: %d", a*b);
    printf("\nDiv is: %d", a/b);
    return 0;
}
```

Output :

```
Enter A and B: 5 6

Sum is: 11
Sub is: -1
Multi is: 30
Div is: 0
```

//2. Write a program to calculate the area and perimeter of a rectangle.

```
#include <stdio.h>
```

```
int main(){
```

```
    int len, bre;
```

```
    printf("Enter Length: ");
```

```
    scanf("%d",&len);
```

```
    printf("Enter Breadth: ");
```

```
    scanf("%d",&bre);
```

```
    printf("\n Area is: %d", len*bre );
```

```
    printf("\n Parameter is: %d", 2*(len+bre) );
```

```
    return 0;
```

```
}
```

OUTPUT:

```
Enter Length: 6
Enter Breadth: 5

Area is: 30
Parameter is: 22
```

// 3. Write a program to calculate the simple interest using arithmetic operators.

```
int main(){

    float P,R,T;

    printf("Enter Principle: ");

    scanf("%f",&P);

    printf("Enter Rate: ");

    scanf("%f",&R);

    printf("Enter Time: ");

    scanf("%f",&T);


    float SI = (P*R*T)/100;

    printf("\n Simple Intrest is: %.2f", SI );


    return 0;

}
```

OUTPUT:

```
Enter Principle: 5000
Enter Rate: 2
Enter Time: 3

Simple Intrest is: 300.00
```

// 4. Write a program to input a 3-digit number and find the sum of its digits.

```
# include <stdio.h>
```

```
int main(){

    int a,b,c,n;

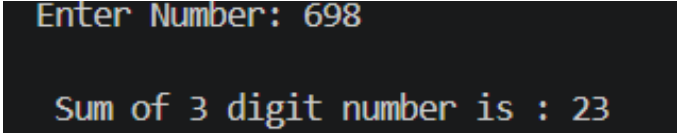
    printf("Enter Number: ");
```

```
scanf("%d",&n);

a = n%10;
n = n/10;
b = n%10;
n = n/10;
c = n%10;
n = n/10;
int sum = a+b+c;
printf("\n Sum of 3 digit number is : %d", sum );

return 0;
}
```

OUTPUT:

A screenshot of a terminal window with a black background and yellow text. The first line shows the prompt 'Enter Number: 698'. The second line shows the output 'Sum of 3 digit number is : 23'.

```
Enter Number: 698
Sum of 3 digit number is : 23
```

// 5. Write a program to convert seconds into hours, minutes, and seconds.

```
# include <stdio.h>
```

```
int main(){
```

```
    int hours, sec, min, cal;
```

```
    printf("Enter Number: ");
```

```
    scanf("%d",&sec);
```

```
    hours = sec/3600;
```

```
    sec = sec%3600;
```

```
    min = sec/60;
```

```
    sec = sec%60;
```

```
    printf("\n Hours: %d", hours );
```

```
printf("\n Minutes: %d", min );
```

```
printf("\n Seconds: %d", sec );
```

```
return 0;
```

```
}
```

OUTPUT:

```
Enter Number: 6000
```

```
Hours: 1
```

```
Minutes: 40
```

```
Seconds: 0
```

2. Relational Operators <>, <=, >=, == & !=

// 1. Write a program to input two numbers and check which number is greater.

```
# include <stdio.h>
```

```
int main() {
```

```
    int num1, num2;
```

```
    printf("Enter two numbers: ");
```

```
    scanf("%d %d", &num1, &num2);
```

```
    if (num1 > num2) {
```

```
        printf("%d is greater than %d\n", num1, num2);
```

```
    } else if (num2 > num1) {
```

```
        printf("%d is greater than %d\n", num2, num1);
```

```
    } else {
```

```
        printf("Both numbers are equal.\n");
```

```
    }
```

```
    return 0;
```

```
}
```

OUTPUT:

```
Enter two numbers: 50 60
```

```
60 is greater than 50
```

// 2. Write a program to check whether a number is equal to zero, positive, or negative.

```
#include <stdio.h>
```

```
int main() {  
    int num1, num2;  
    printf("Enter a numbers: ");  
    scanf("%d", &num1);  
  
    if (num1 > 0) {  
        printf("%d is a positive number.\n", num1);  
    } else if (num1 < 0) {  
        printf("%d is not a positive number.\n", num1);  
    } else {  
        printf("%d is zero.\n", num1);  
    }  
  
    return 0;  
}
```

OUTPUT:

```
Enter a numbers: 50  
50 is a positive number.
```

```
Enter a numbers: -50  
-50 is not a positive number.
```

// 3. Write a program to input a student's marks and check whether the student has passed or failed.

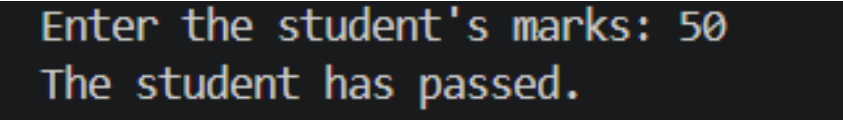
```
#include <stdio.h>
```

```
int main() {  
    int marks;  
    printf("Enter the student's marks: ");  
    scanf("%d", &marks);  
  
    if (marks >= 40) {
```

```
    printf("The student has passed.\n");
} else {
    printf("The student has failed.\n");
}

return 0;
}
```

OUTPUT:



```
Enter the student's marks: 50
The student has passed.
```

//4. Write a program to find the largest of two numbers using relational operators.

```
#include <stdio.h>
```

```
int main() {
    int num1, num2;

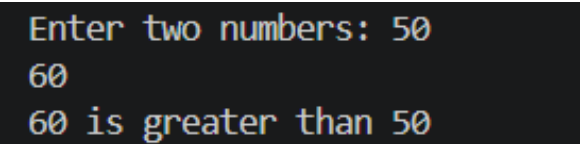
    printf("Enter two numbers: ");

    scanf("%d %d", &num1, &num2);

    if (num1 > num2) {
        printf("%d is greater than %d\n", num1, num2);
    } else if (num2 > num1) {
        printf("%d is greater than %d\n", num2, num1);
    } else {
        printf("Both numbers are equal.\n");
    }

    return 0;
}
```

OUTPUT:



```
Enter two numbers: 50
60
60 is greater than 50
```

//5. Write a program to check whether a given year is a leap year.

```
#include <stdio.h>
```

```
int main() {  
    int year;  
    printf("Enter a year: ");  
    scanf("%d", &year);  
  
    if ((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0)) {  
        printf("%d is a leap year.\n", year);  
    } else {  
        printf("%d is not a leap year.\n", year);  
    }  
  
    return 0;  
}
```

OUTPUT:

```
Enter a year: 2006  
2006 is not a leap year.
```

3. Logical Operators &&, ||, !

//1. Write a program to check whether a number lies between 10 and 50.

```
#include <stdio.h>
```

```
int main() {
```

```
    int num;
```

```
    printf("Enter a number: ");
```

```
    scanf("%d", &num);
```

```
    if (num > 10 && num < 50) {
```

```
        printf("%d lies between 10 and 50.\n", num);
```

```
    } else {
```

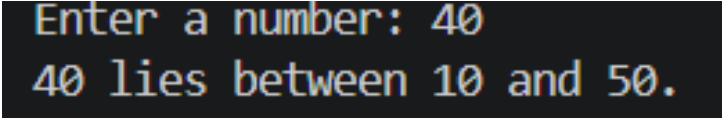
```
        printf("%d does not lie between 10 and 50.\n", num);
```

```
    }
```

```
    return 0;
```

```
}
```

OUTPUT:



```
Enter a number: 40
40 lies between 10 and 50.
```

//2. Write a program to check whether a person is eligible to vote based on age.

```
#include <stdio.h>
```

```
int main() {
```

```
    int age;
```

```
    printf("Enter your age: ");
```

```
    scanf("%d", &age);
```

```
    if (age >= 18) {
```

```
        printf("You are eligible to vote.\n");
```

```
    } else {
```

```
        printf("You are not eligible to vote.\n");
```

```
    }
```

```
    return 0;
```

```
}
```

OUTPUT:

```
Enter your age: 50
You are eligible to vote.
```

//3. Write a program to check whether a number is divisible by both 3 and 5.

```
#include <stdio.h>
```

```
int main() {
```

```
    int num;
```

```
    printf("Enter a number: ");
```

```
    scanf("%d", &num);
```

```
    if (num % 3 == 0 && num % 5 == 0) {
```

```
        printf("%d is divisible by both 3 and 5.\n", num);
```

```
    } else {
```

```
        printf("%d is not divisible by both 3 and 5.\n", num);
```

```
    }
```

```
    return 0;
```

```
}
```

OUTPUT:

```
Enter a number: 60
60 is divisible by both 3 and 5.
```

//4. Write a program to check whether a number is positive and even.

```
#include <stdio.h>
```

```
int main() {
```

```
    int num;
```

```
    printf("Enter a number: ");
```

```
    scanf("%d", &num);
```

```
    if (num > 0 ) {
```

```
        if (num % 2 == 0) {
```

```
            printf("%d is a positive and even number.\n", num);
```

```
        } else {
```

```

        printf("%d is a positive but not an even number.\n", num);
    }
} else {
    printf("%d is not a positive number.\n", num);
}
return 0;
}

```

OUTPUT:

```

Enter a number: 31
31 is a positive but not an even number.

```

//5. Write a program to check whether a character is a vowel or consonant using logical operators.

```

#include <stdio.h>

int main() {
    char ch;

    printf("Enter a character: ");

    scanf(" %c", &ch);

    if ((ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u') ||
        (ch == 'A' || ch == 'E' || ch == 'I' || ch == 'O' || ch == 'U')) {
        printf("%c is a vowel.\n", ch);
    } else {
        printf("%c is a consonant.\n", ch);
    }

    return 0;
}

```

OUTPUT:

```

Enter a character: v
v is a consonant.

```

```

Enter a character: A
A is a vowel.

```

4. Assignment Operators = += -= *= /= %=

//1. Write a program to demonstrate all compound assignment operators.

```
#include <stdio.h>
```

```
int main() {
```

```
    int a, b;
```

```
    printf("Enter two numbers: ");
```

```
    scanf("%d %d", &a, &b);
```

```
    printf("\nInitial values: a = %d, b = %d\n", a, b);
```

```
    a += b; // Addition assignment
```

```
    printf("After a += b: a = %d\n", a);
```

```
    a -= b; // Subtraction assignment
```

```
    printf("After a -= b: a = %d\n", a);
```

```
    a *= b; // Multiplication assignment
```

```
    printf("After a *= b: a = %d\n", a);
```

```
    if (b != 0) {
```

```
        a /= b; // Division assignment
```

```
        printf("After a /= b: a = %d\n", a);
```

```
    } else {
```

```
        printf("Division by zero is not allowed.\n");
```

```
    }
```

```
    if (b != 0) {
```

```
        a %= b; // Modulus assignment
```

```
        printf("After a %%= b: a = %d\n", a);
```

```

    } else {
        printf("Modulus by zero is not allowed.\n");
    }

    return 0;
}

```

OUTPUT:

```

Enter two numbers: 50
40

Initial values: a = 50, b = 40
After a += b: a = 90
After a -= b: a = 50
After a *= b: a = 2000
After a /= b: a = 50
After a %= b: a = 10

```

//2. Write a program that increases a number by 10 using +=.

```

#include <stdio.h>

int main() {
    int num;

    printf("Enter a number: ");

    scanf("%d", &num);

    num += 10;

    printf("The number after increasing by 10 is: %d\n", num);

    return 0;
}

```

OUTPUT:

```

Enter a number: 50
The number after increasing by 10 is: 60

```

//3. Write a program that decreases a number by 5 using -=.

```

#include <stdio.h>

int main() {
    int num;

    printf("Enter a number: ");

    scanf("%d", &num);

```

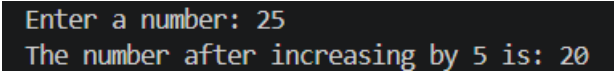
```
num -= 5;

printf("The number after increasing by 10 is: %d\n", num);

return 0;

}
```

OUTPUT:



```
Enter a number: 25
The number after increasing by 5 is: 20
```

//4. Write a program to calculate the final value of a variable after applying +=, -=, *=, and /=.

```
# include<stdio.h>
```

```
int main(){

    int num;

    printf("Enter a number: ");

    scanf("%d", &num);

    num += 10; // Increase by 10

    printf("After += 10: %d\n", num);

    num -= 5; // Decrease by 5

    printf("After -= 5: %d\n", num);

    num *= 2; // Multiply by 2

    printf("After *= 2: %d\n", num);

    if (num != 0) {

        num /= 3; // Divide by 3

        printf("After /= 3: %d\n", num);

    } else {

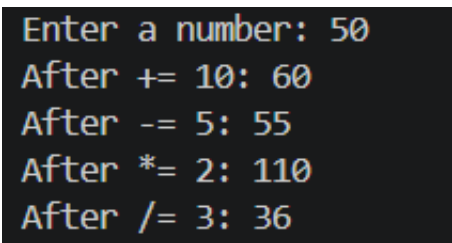
        printf("Division by zero is not allowed.\n");

    }

    return 0;

}
```

OUTPUT:



```
Enter a number: 50
After += 10: 60
After -= 5: 55
After *= 2: 110
After /= 3: 36
```

5. Bitwise Operators.

//1. Write a C program to find the result of bitwise AND of two numbers.

```
#include <stdio.h>
```

```
int main() {
```

```
    int num1, num2;
```

```
    printf("Enter two numbers: ");
```

```
    scanf("%d %d", &num1, &num2);
```

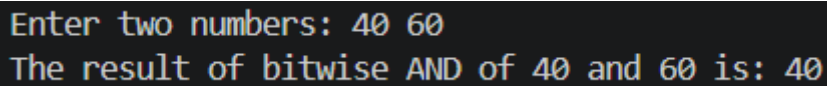
```
    int result = num1 & num2;
```

```
    printf("The result of bitwise AND of %d and %d is: %d\n", num1, num2, result);
```

```
    return 0;
```

```
}
```

OUTPUT:



```
Enter two numbers: 40 60
The result of bitwise AND of 40 and 60 is: 40
```

//2. Write a program to demonstrate AND, OR, and XOR operators.

```
#include <stdio.h>
```

```
int main() {
```

```
    int num1, num2;
```

```
    printf("Enter two numbers: ");
```

```
    scanf("%d %d", &num1, &num2);
```

```
    int and_result = num1 & num2;
```

```
    int or_result = num1 | num2;
```

```
    int xor_result = num1 ^ num2;
```

```
    printf("The result of bitwise AND of %d and %d is: %d\n", num1, num2, and_result);
```

```
    printf("The result of bitwise OR of %d and %d is: %d\n", num1, num2, or_result);
```

```
printf("The result of bitwise XOR of %d and %d is: %d\n", num1, num2, xor_result);
```

```
return 0;
```

```
}
```

OUTPUT:

```
Enter two numbers: 50 60
The result of bitwise AND of 50 and 60 is: 48
The result of bitwise OR of 50 and 60 is: 62
The result of bitwise XOR of 50 and 60 is: 14
```

//3. Write a program to check whether a number is even or odd using bitwise AND.

```
#include <stdio.h>
```

```
int main() {
```

```
    int num;
```

```
    printf("Enter a number: ");
```

```
    scanf("%d", &num);
```

```
    if (num & 1) {
```

```
        printf("%d is an odd number.\n", num);
```

```
    } else {
```

```
        printf("%d is an even number.\n", num);
```

```
    }
```

```
    return 0;
```

```
}
```

OUTPUT:

```
Enter a number: 53
53 is an odd number.
```

//4. Write a program to multiply a number by 2 using the left shift operator.

```
#include <stdio.h>
```

```
int main() {
```

```
    int num;
```

```
    printf("Enter a number: ");
```

```
scanf("%d", &num);
```

```
int result = num << 1;
```

```
printf("The result of multiplying %d by 2 using left shift is: %d\n", num, result);
```

```
return 0;
```

```
}
```

OUTPUT:

```
Enter a number: 6
The result of multiplying 6 by 2 using left shift is: 12
```

//5. Write a program to divide a number by 2 using the right shift operator.

```
#include <stdio.h>
```

```
int main() {
```

```
    int num;
```

```
    printf("Enter a number: ");
```

```
    scanf("%d", &num);
```

```
    int result = num >> 1;
```

```
    printf("The result of dividing %d by 2 using right shift is: %d\n", num, result);
```

```
    return 0;
```

```
}
```

OUTPUT:

```
Enter a number: 6
The result of dividing 6 by 2 using right shift is: 3
```

6. Conditional Operator.

//1. Write a program to find the largest of two numbers using the conditional operator.

```
#include<stdio.h>
int main(){
    int a,b;
    printf("Enter A and B: ");
    scanf("%d %d", &a, &b);

    (a>b)? printf("\n A is largest: %d", a): printf("\n B is largest: %d", b);
    return 0;
}
```

OUTPUT:

```
Enter A and B: 6 8

B is largest: 8
```

//2. Write a program to check whether a number is even or odd using ?:.

```
#include<stdio.h>
```

```
int main(){
```

```
    int n;
```

```
    printf("Enter Number: ");
```

```
    scanf("%d", &n);
```

```
    (n%2==0)? printf("\n Your Number is Even: %d", n ): printf("\n Your Number is Odd: %d", n );
```

```
    return 0;
```

```
}
```

OUTPUT:

```
Enter Number: 62

Your Number is Even: 62
```

//3. Write a program to find the smallest of two numbers using ?:.

```
#include <stdio.h>
```

```
int main(){
```

```
    int a,b;
```

```
    printf("Enter A and B: ");
```

```
scanf("%d %d", &a, &b);
```

```
(a<b)? printf("\n A is smallest: %d", a ): printf("\n B is smallest: %d", b );
```

```
return 0;
```

```
}
```

OUTPUT:

```
Enter A and B: 40 60
```

```
A is smallest: 40
```

// 4. Write a program to check whether a person is eligible or not eligible to vote using ?:.

```
#include <stdio.h>
```

```
int main(){
```

```
    int age;
```

```
    printf("Enter Age: ");
```

```
    scanf("%d",&age);
```

```
    (age>=18)? printf("\n You are Eligible for Voting: %d", age ): printf("\n You are Not Eligible for Voting: %d", age );
```

```
return 0;
```

```
}
```

OUTPUT:

```
Enter Age: 50
```

```
You are Eligible for Voting: 50
```

//5. Write a program to find the largest of three numbers using nested conditional operators in ternary operators

```
#include<stdio.h>
```

```
int main(){
```

```
    int a,b,c;
```

```
    printf("Enter A, B and C: ");
```

```
scanf("%d %d %d", &a, &b, &c);
```

```
(a>b)? (a>c? printf("\n A is largest: %d", a): printf("\n C is largest: %d", c)): (b>c? printf("\n B is largest: %d", b): printf("\n C is largest: %d", c));
```

```
return 0;
```

```
}
```

OUTPUT:

```
Enter A, B and C: 6 8 30
C is largest: 30
```

7. Increment and Decrement++ --

//1. Write a program to demonstrate pre-increment and post-increment.

```
#include <stdio.h>
```

```
int main() {
```

```
    int num;
```

```
    printf("Enter a number: ");
```

```
    scanf("%d", &num);
```

```
    int pre_increment = ++num;
```

```
    int post_increment = num++;
```

```
    printf("After pre-increment, the value is: %d\n", pre_increment);
```

```
    printf("After post-increment, the value is: %d\n", post_increment);
```

```
    printf("The final value of num is: %d\n", num);
```

```
    return 0;
```

```
}
```

OUTPUT:

```
Enter a number: 50
After pre-increment, the value is: 51
After post-increment, the value is: 51
```

//2. Write a program to demonstrate pre-decrement and post-decrement.

```
#include <stdio.h>
```

```
int main() {
```

```
    int num;
```

```
    printf("Enter a number: ");
```

```
    scanf("%d", &num);
```

```
    int pre_decrement = --num;
```

```
    int post_decrement = num--;
```

```
    printf("After pre-decrement, the value is: %d\n", pre_decrement);
```

```
    printf("After post-decrement, the value is: %d\n", post_decrement);
```

```
    printf("The final value of num is: %d\n", num);
```

```
    return 0;
```

```
}
```

OUTPUT:

```
Enter a number: 6
After pre-decrement, the value is: 5
After post-decrement, the value is: 5
The final value of num is: 4
```

8. Special Operators

//1. Important special operators in C include sizeof, comma.

```
#include <stdio.h>
```

```
int main() {
```

```
    int num;
```

```
    printf("Enter a number: ");
```

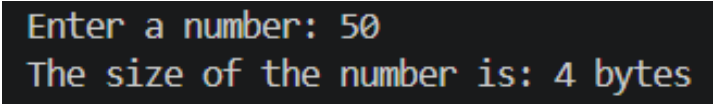
```
    scanf("%d", &num);
```

```
    printf("The size of the number is: %zu bytes", sizeof(num));
```

```
    return 0;
```

```
}
```

OUTPUT:

A screenshot of a terminal window with a dark background. It shows the output of the first program: "Enter a number: 50" followed by "The size of the number is: 4 bytes".

```
Enter a number: 50
The size of the number is: 4 bytes
```

//2. Write a program to find the size of int, float, char, and double using sizeof.

```
#include <stdio.h>
```

```
int main() {
```

```
    int num;
```

```
    float fnum;
```

```
    char ch;
```

```
    double dnum;
```

```
    printf("Enter a number: ");
```

```
    scanf("%d", &num);
```

```
    printf("Enter a float number: ");
```

```
    scanf("%f", &fnum);
```

```
    printf("Enter a character: ");
```

```
    scanf(" %c", &ch);
```

```

printf("Enter a double number: ");
scanf("%lf", &dnum);

printf("The size of int is: %zu bytes\n", sizeof(num));
printf("The size of float is: %zu bytes\n", sizeof(fnum));
printf("The size of char is: %zu bytes\n", sizeof(ch));
printf("The size of double is: %zu bytes\n", sizeof(dnum));

return 0;
}

```

OUTPUT:

```

Enter a number: 50
Enter a float number: 5.5
Enter a character: k
Enter a double number: 5.055
The size of int is: 4 bytes
The size of float is: 4 bytes
The size of char is: 1 bytes
The size of double is: 8 bytes

```

//3. Write a program to use the comma operator and find the final value of a variable.

```

#include <stdio.h>

int main() {
    int num = 10;

    num = (num += 5, num *= 2, num - 3);

    printf("The final value of num is: %d\n", num);

    return 0;
}

```

OUTPUT:

```

The final value of num is: 27

```

9. OUTPUTS

OUTPUT 1 IS:

26

OUTPUT 2 IS:

11 10

OUTPUT 3 IS:

6 3 8

OUTPUT 4 IS:

1
7
6

OUTPUT 5 IS:

6 6

OUTPUT 6 IS:

0

OUTPUT 7 IS:

6
6
4

OUTPUT 8 IS:

4

OUTPUT 9 IS:

10
2